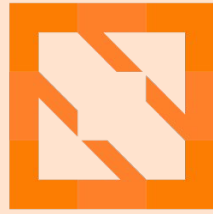




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WELCOME TO VALENCIA





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Scaling K8s Nodes Without Breaking the Bank nor Your Sanity

Brandon Wagner & Nick Tran, AWS



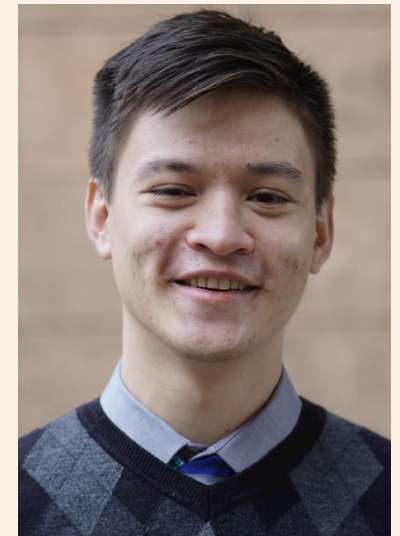
Scaling K8s Nodes Without Breaking the Bank nor Your Sanity

Agenda

- What is Spot? 🐕
- Best Practices
- K8s & Spot ❤️
- Autoscaling your nodes
 - Cluster Autoscaler
 - Karpenter
- Demo



Brandon Wagner
Software Engineer
AWS



Nick Tran
Software Engineer
AWS



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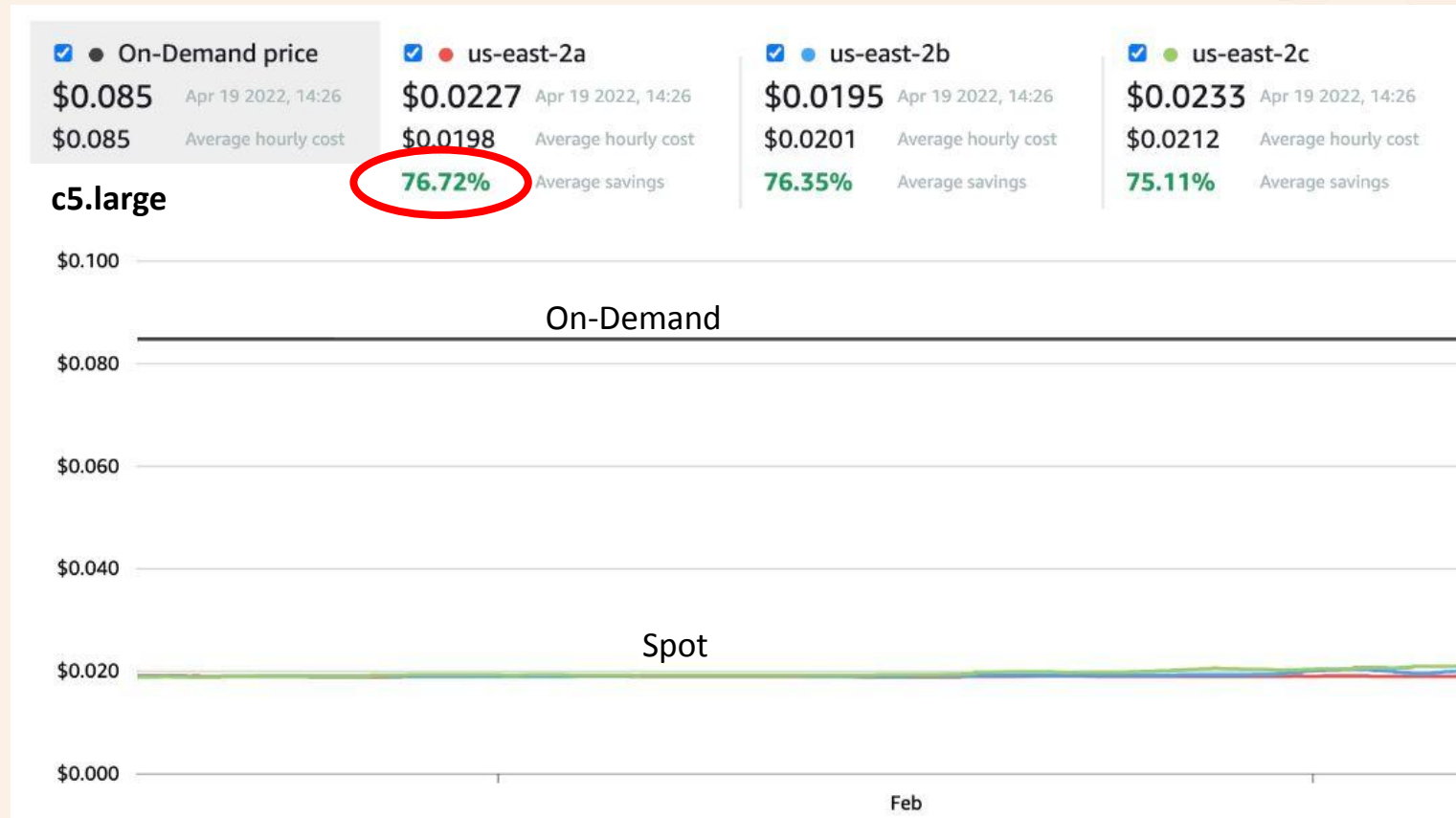


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What is EC2 Spot?

- Spare VM Capacity
- Available at a discount
- Interruptible
 - 2-min notice



Spot Best Practices

- Don't set a Spot max price
- Flexible instance type requests
- Rebalance Recommendations



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Spot Best Practices - Don't Set a Max Price

- Spot pricing model overhaul
- Long-term supply and demand





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Spot Best Practices - Flexible Instance Types

- Increases Spot instance availability
- Capacity pools
- Extend instance runtime
 - w/ capacity-optimized

C5	1a	1b	1c	On-demand
8XL	\$0.28	\$0.27	\$0.29	\$1.76
2XL	\$0.08	\$0.07	\$0.08	\$0.44
L	\$0.01	\$0.01	\$0.04	\$0.11

Example Hourly Prices



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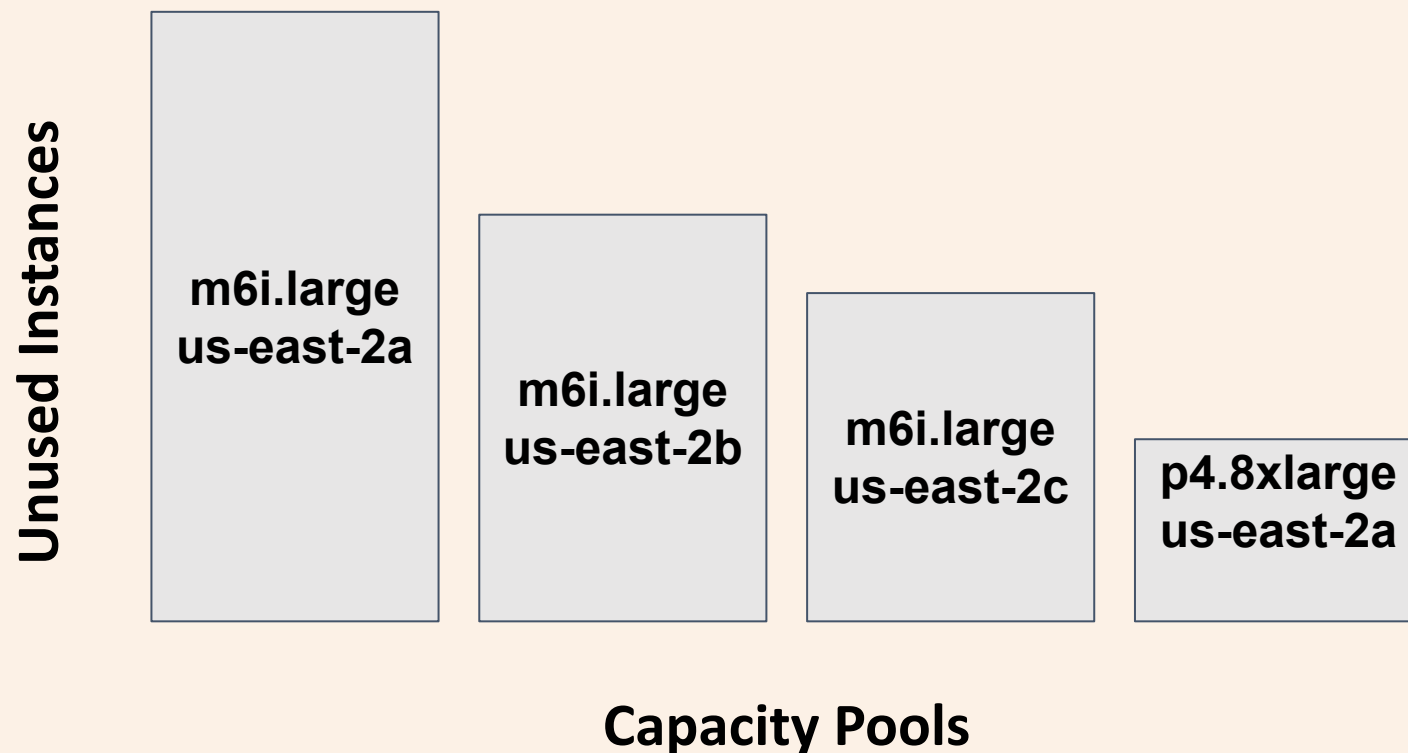


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Spot Best Practices - Flexible Instance Types

- Increases Spot instance availability
- Capacity pools
 - Extend instance runtime
 - w/ capacity-optimized





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Spot Best Practices - Flexible Instance Types

- Increases Spot instance availability
- Capacity pools
- Extend instance runtime
 - w/ capacity-optimized

Instance Type	vCPU	Memory GiB	Savings over On-Demand*	Frequency of interruption
r6g.large	2	16	78%	10-15% ■ □ □ □ □
m4.large	2	8	81%	<5% □ □ □ □ □
c6g.large	2	4	71%	5-10% ■ □ □ □ □
t3.medium	2	4	70%	<5% □ □ □ □ □
im4gn.large	2	8	70%	5-10% ■ □ □ □ □
is4gen.large	2	12	70%	5-10% ■ □ □ □ □
m5ad.large	2	8	81%	<5% □ □ □ □ □
c6i.large	2	4	76%	<5% □ □ □ □ □

<https://aws.amazon.com/ec2/spot/instance-advisor>



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Spot Best Practices - Rebalance Recommendations

- Early warning to indicate a possible Spot interruption
- More time to gracefully shutdown workloads





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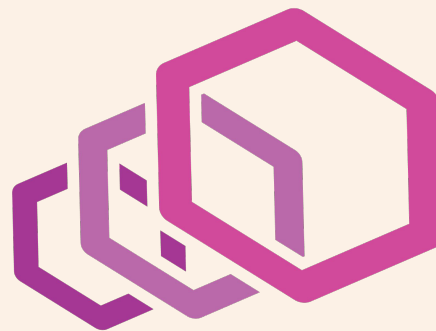
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Common Workloads

- Quick Continuous Integration
- Batch processing
- Stateless APIs



TEKTON



envoy



Istio



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K8s and Spot

- github.com/aws/aws-node-termination-handler
 - Interruption Termination Notifications
 - Rebalance Recommendations
- Pod Disruption Budgets (PDBs)



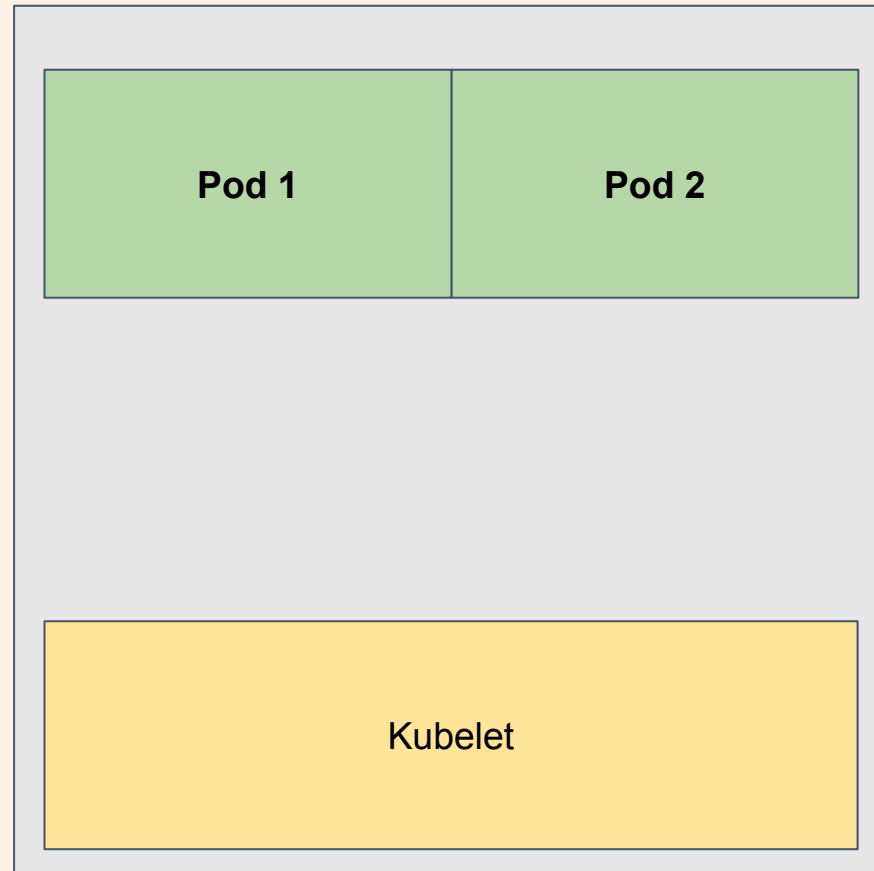
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K8s and Spot - K8s Eviction API





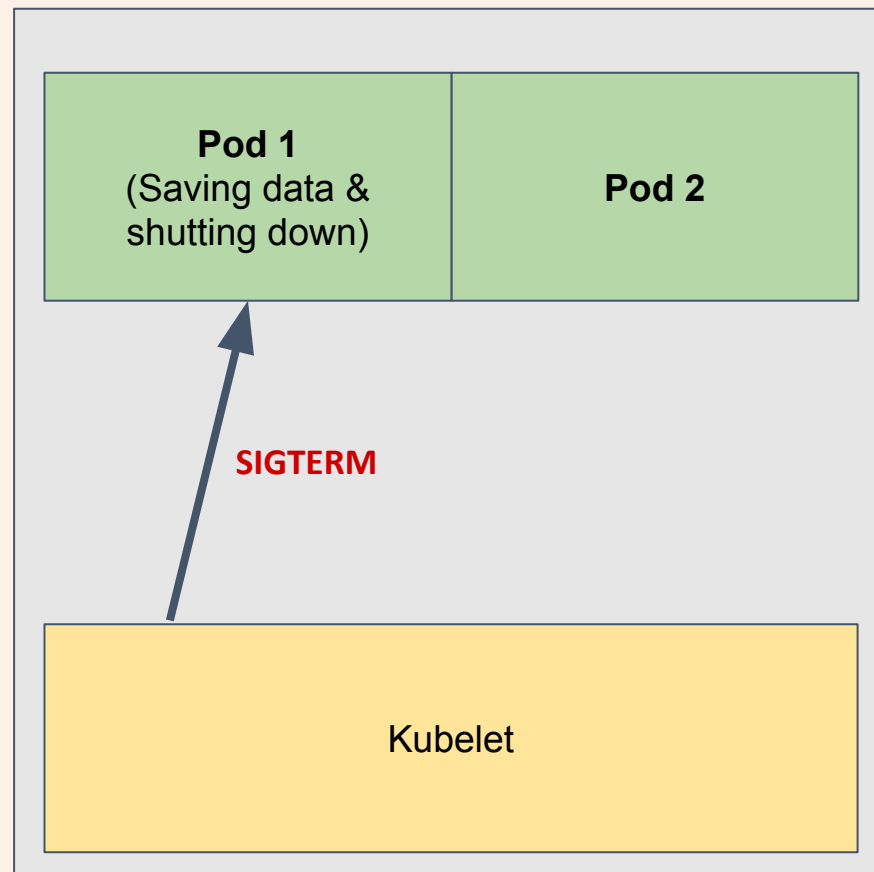
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K8s and Spot - K8s Eviction API





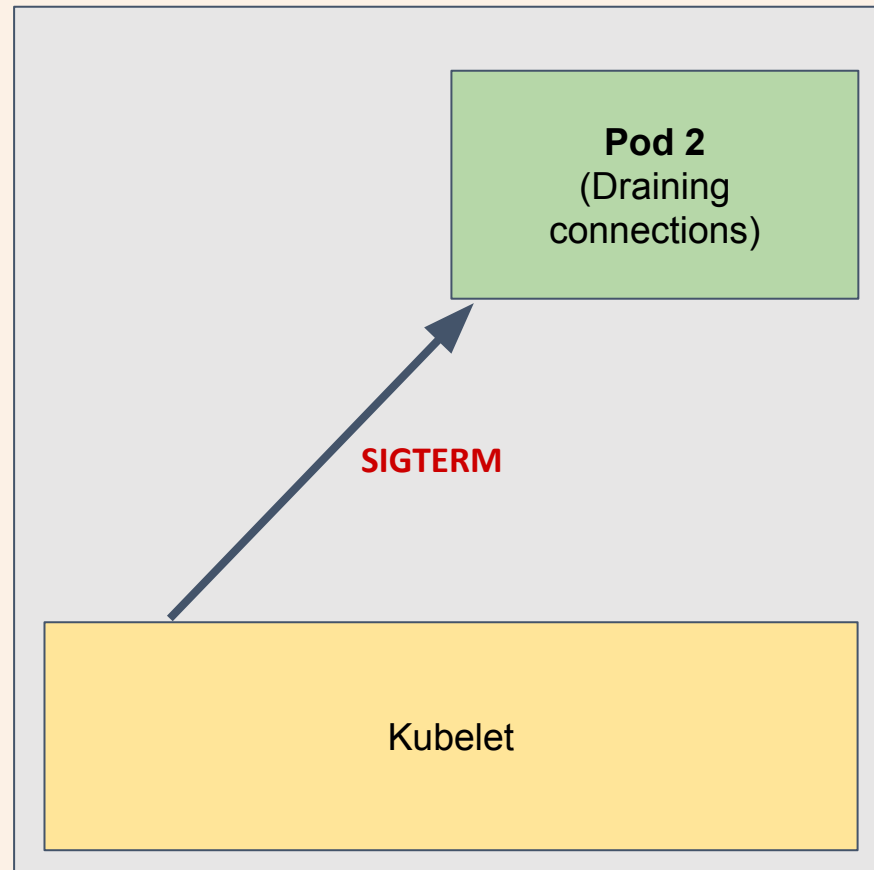
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K8s and Spot - K8s Eviction API



**A FEW
MINUTES
LATER...**



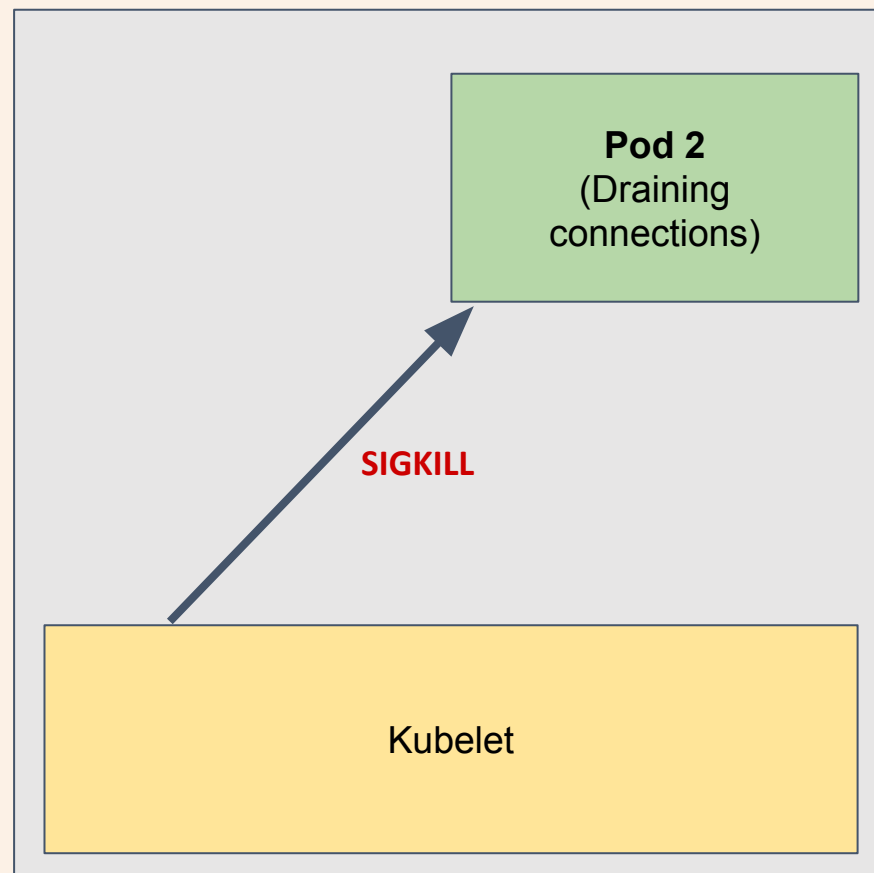
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K8s and Spot - K8s Eviction API



K8s and Spot - K8s Eviction API

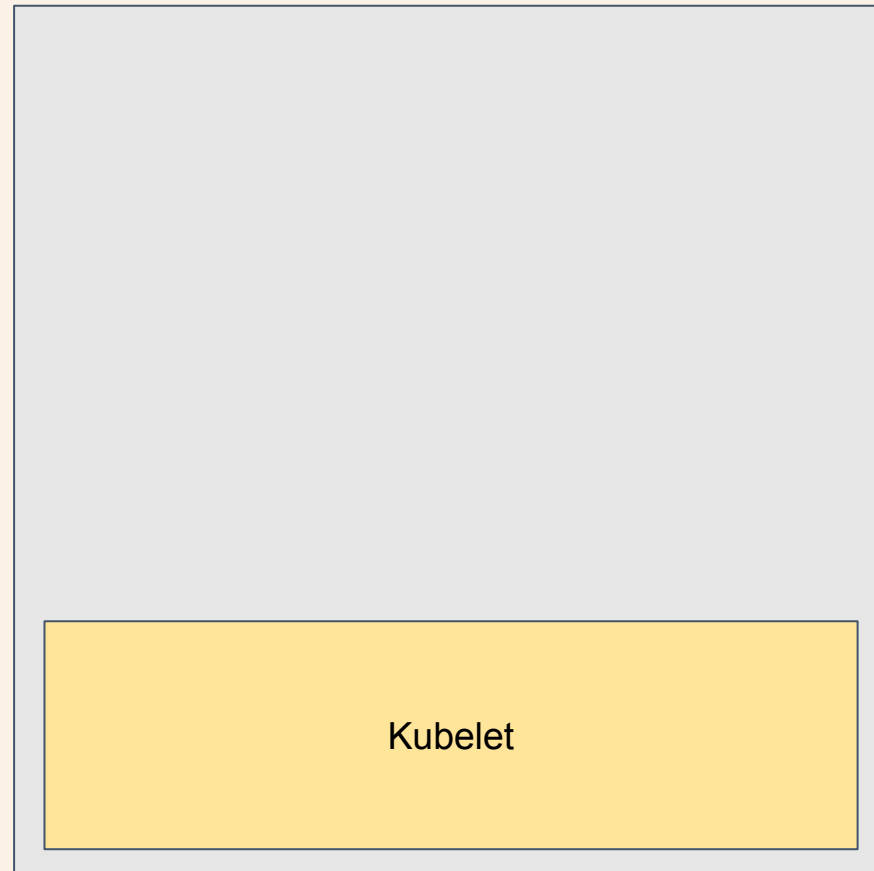


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Autoscaling your Cluster

- Pod Autoscaling
 - Horizontal Pod Autoscaler (HPA)
 - Vertical Pod Autoscaler (VPA)
- Node Autoscaling
 - Cluster Autoscaler
 - Karpenter



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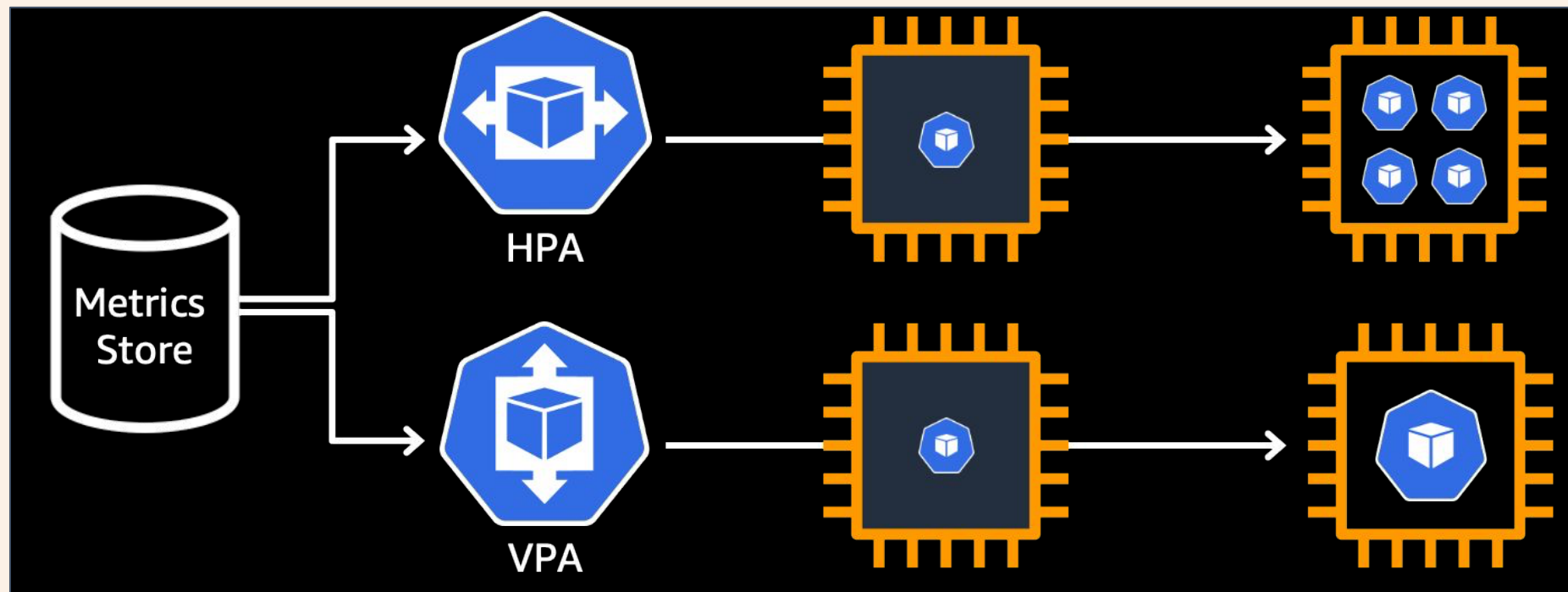


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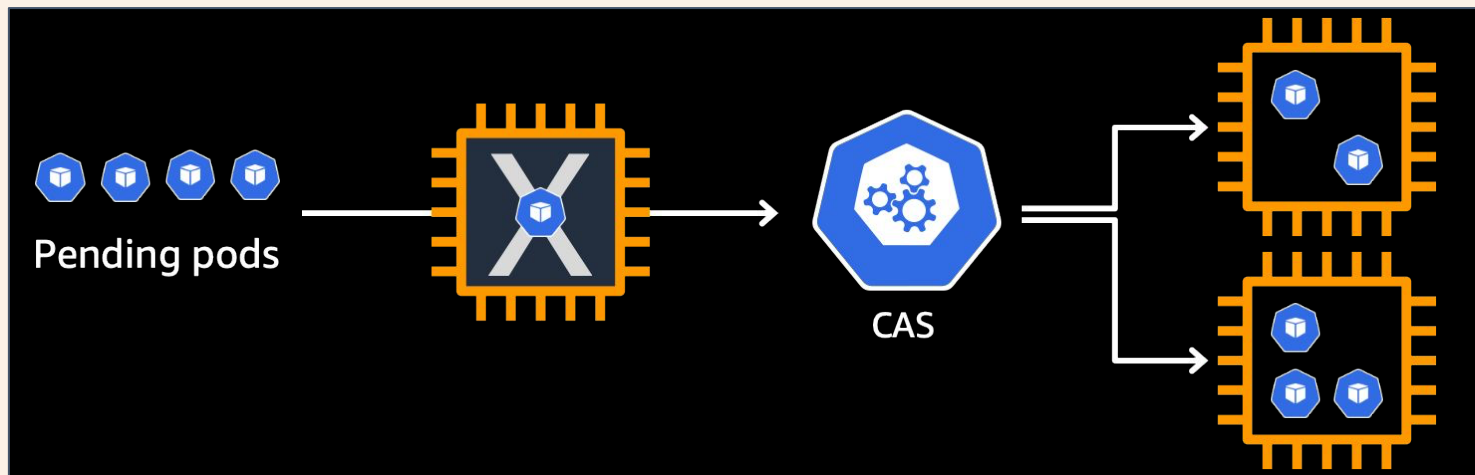
HPA & VPA

- Horizontally scale: adjust pod replicas
- Vertically scale: adjust resources of pods



Cluster Autoscaler

- Simple interface between EC2 AutoScaling Groups (ASGs)
- Increments desired capacity in response to pending pods
- Need to create resource workloads per type of pod resource request





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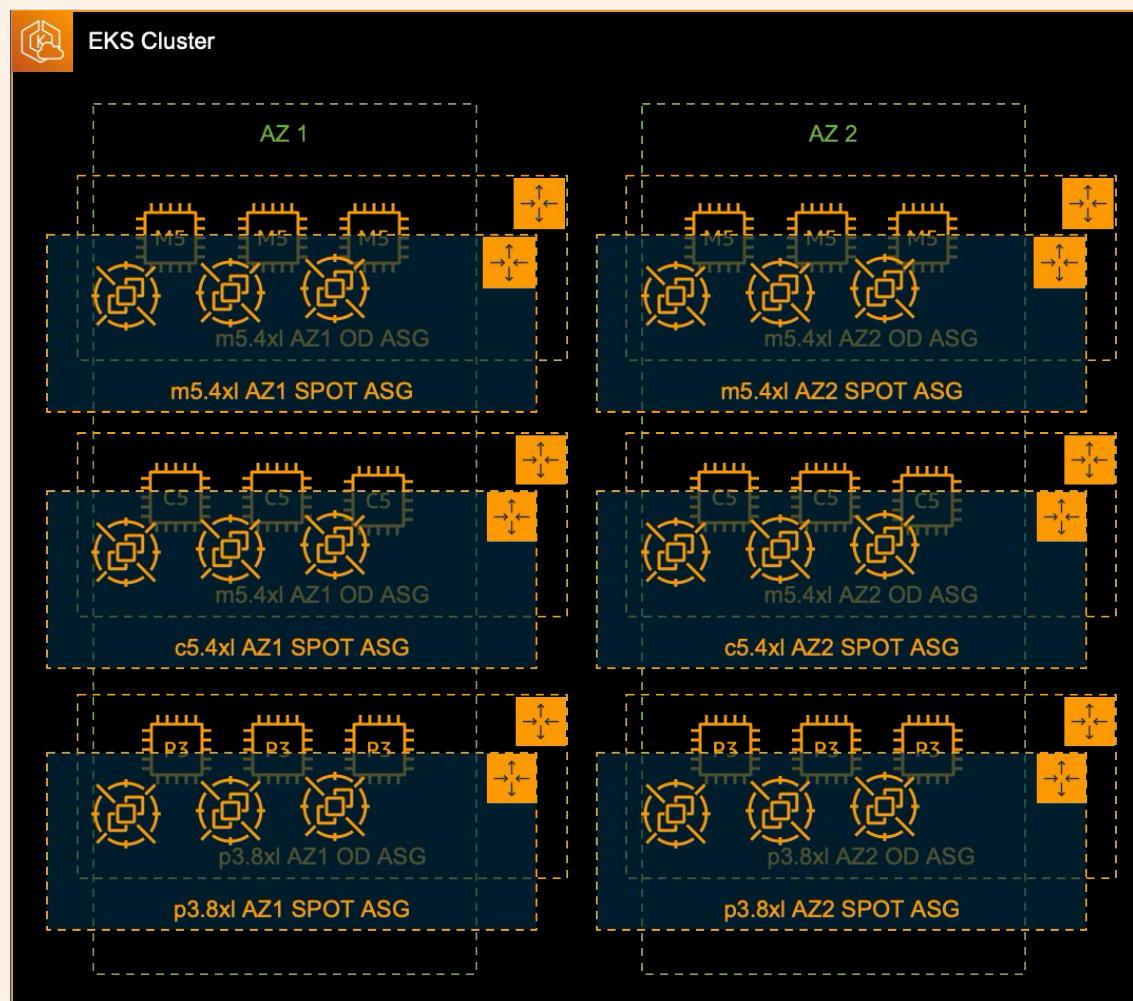


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Cluster Autoscaler

- Externally Managed Infrastructure
- [Spot, OD] x [AZ1, AZ2] x [m5, c5, p3] = 12 ASGs





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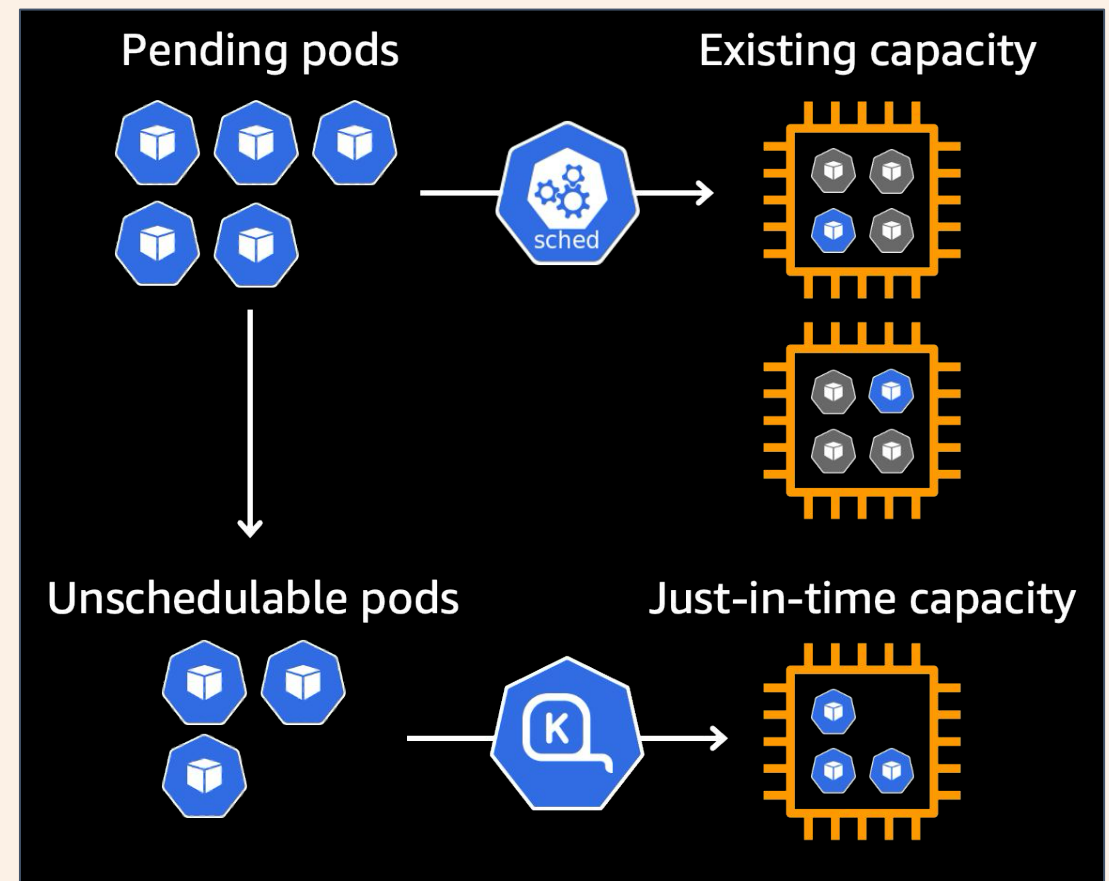


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Karpenter

- Groupless Node Autoscaler
- Just-in-Time Provisioning
 - Pending Pods
- github.com/aws/karpenter
 - Vendor neutral cloud provider interface





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Karpenter

- Provisioner CRD
- Requirements
 - Scheduling Constraints
 - Well Known Labels
 - Capacity Type
- Cloud Provider

```
apiVersion: karpenter.sh/v1alpha5
kind: Provisioner
metadata:
  name: default
spec:
  ttlSecondsAfterEmpty: 60
  ttlSecondsUntilExpired: 525600 # ~6 days
  requirements:
    - key: kubernetes.io/arch
      operator: In
      values:
        - arm64
        - amd64
    - key: karpenter.sh/capacity-type
      operator: In
      values:
        - spot
        - on-demand
  provider:
    kind: AWS
    securityGroupSelector:
      karpenter.sh/discovery: my-cluster
    subnetSelector:
      karpenter.sh/discovery: my-cluster
    instanceProfile: 'KarpenterNodeInstanceProfile-my-cluster'
```



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Karpenter

- Flexibility
 - CPU Architecture

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apiVersion: karpenter.sh/v1alpha5
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Karpenter

- Flexibility
 - CPU Architecture
 - Capacity Type

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Karpenter

Where are the Instance Types?

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Karpenter

Where are the Instance Types?

```
spec:
  containers:
    - image: pause
      name: gpu-pod
  resources:
    limits:
      nvidia.com/gpu: 1
```

```
apiVersion: karpenter.sh/v1alpha5
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```

Karpenter - AWS Cloud Provider

- EC2 Fleet API
 - Flexible to many instance types
 - Chooses optimal AZ and instance type
- Spot to On-Demand Fallback



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Autoscaling Nodes - Karpenter

- Scaling down
 - ttlSecondsAfterEmpty
 - ttlSecondsUntilExpired
- Follows Graceful Node Shutdown

```
apiVersion: karpenter.sh/v1alpha5
kind: Provisioner
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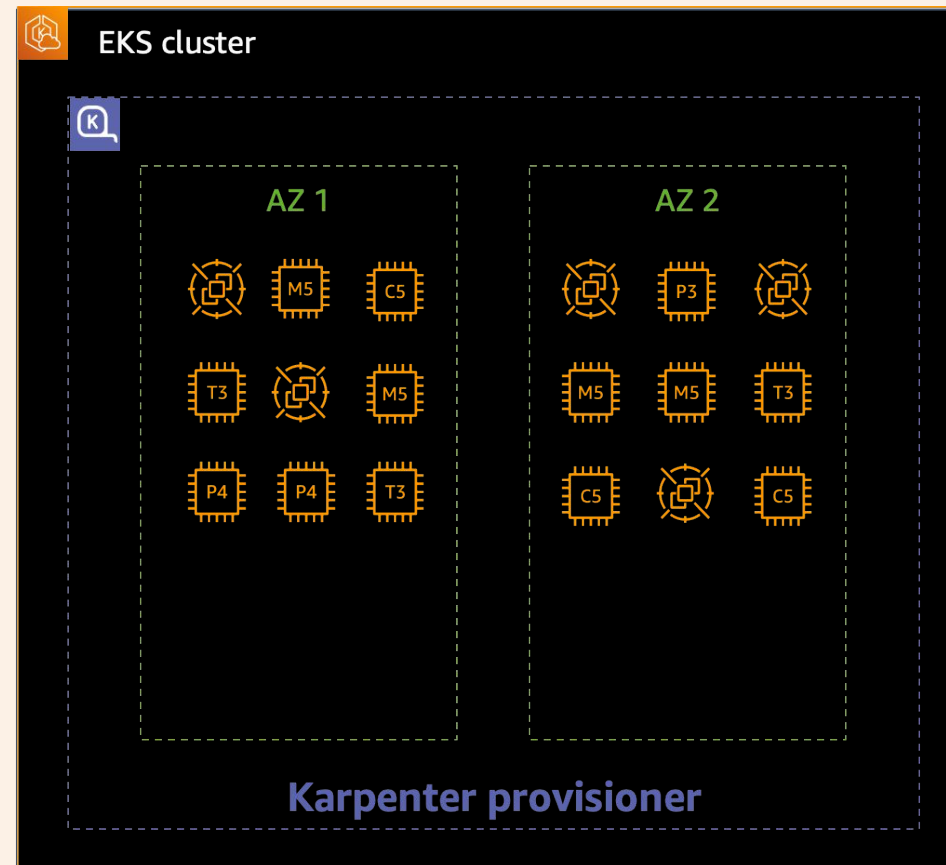


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Autoscaling Nodes - CAS vs Karpenter

- [Spot, OD] x [AZ1, AZ2] x [m5, c5, p4, t3] = 16 ASGs
- One provisioner!



Wrapping Up

- Spot Best Practices
- K8s and Spot!
- Autoscaling nodes with Cluster Autoscaler and Karpenter

Demo!

- Provisioners
- Stuff
- More stuff